

Successful Planning

Aerothermal Interaction TMMV59 – Assignment 2

Planning report for a CFD-related engineering project.

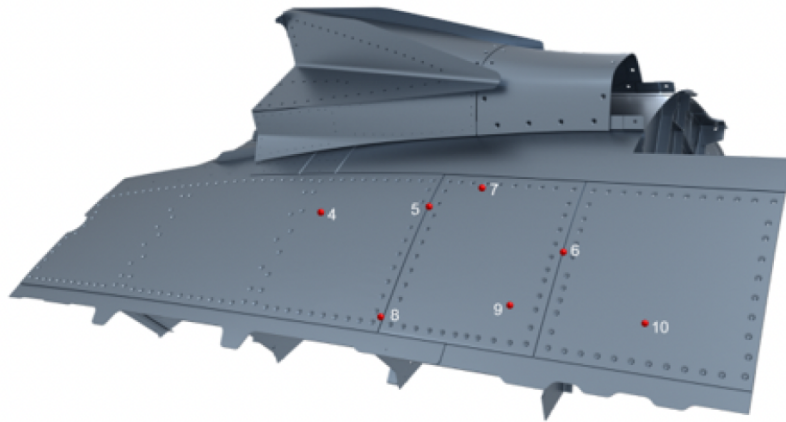
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1 Introduction

1.1 Background and Problem Statement

The study of heat is of great importance in the field of aeronautics. It can significantly affect structural integrity of the aircraft fuselage, having an impact on performance parameters and safety. Prediction of heat can allow for more optimal design choices.

Plane&Simple AB wants to create a method for investigating the heat coming from the engine exhaust and propagating through the hull of the plane. One way to study the heat is through experimentation, however flight tests can be both expensive and time consuming, and the ambition is therefore to create a computational model to predict the heat. By utilizing conjugate heat transfer, both solid and air flow heat transfer may be taken into account. In this way the real flight conditions may be mimicked while avoiding expensive experiments.

Currently Plane&Simple have no such numerical model that takes into account both conduction and convection.

1.2 Aim

The aim of this study is to investigate the impact of engine exhaust heat on the surface of a plane fuselage. To accomplish this a conjugate heat transfer model will be created using computational fluid dynamics. The following research questions are thus posed.

- What is the method of creating a reliable solid simulation model using experimental data to predict temperature fields on the fuselage surface in proximity to the engine?
- Utilizing a CHT simulation approach, what is the impact on the temperature field when using steady versus unsteady turbulence modeling?

1.3 Deliverables

The project deliverables include an experimentally validated computational model for predicting solid heat transfer. Furthermore a CHT model coupling the conduction and convection should be delivered along with the methodology employed explaining necessary steps. Both models must be able to output the temperature field and heat flux on the plane surface for review.

1.4 Delimitations

The study shall not cover the calculation of strains and stresses which arises in the aircraft structure due to heat. In addition, a full scale model will not be created, only a section of the aircraft in proximity to the exhaust gases will be modeled.

2 Methodology

This chapter describes how the work is structured. Included is organization of the experiments, model creation as well as plans for verification and validation of the model.

2.1 Organization of experimental testing

The laboratory campaign must provide the measurements needed to calibrate the contact thermal resistances in the solid model and also evaluate its predictions using a separate test. For each test, the heating conditions must be known, and temperature changes at various locations on the component must be measured. So the heating regimen to apply, the measurements to get and the data reserved for validation have to be specified before testing. Laboratory access is limited to six days during the project’s first five weeks. The schedule we propose involves using three days to set up the test rig, two days to conduct the tests, and one additional day reserved for repeating a test if a problem arise. Post-processing of data can be carried out outside the laboratory. The experiment schedule is further detailed in Tab. 1.

Table 1: Proposed scheduling for experiment testing

Time frame	Required action
Week 1-2: Protocol preparation	Book the six days of laboratory time, obtain the specifications for the setup and equipment, and define the tests, measurements, and durations.
Week 3, day 1-3: Laboratory preparation	Install the fuselage part, the heating source, and the thermocouples. Record their positions, check the sensors and the recording system, then verify operation of the heating system.
Week 3, day 4-5: Conducting tests	Carry out the scheduled sequences, record the temperatures and heating conditions, and then verify the files after each test.
Week 4, day 1: Lab catch-up day	If necessary, correct any detected problem and repeat the relevant test.
Week 4, day 2-5: Post-processing	Organize files, associate each measurement with its sensor and timestamp, plot the temperature curves, and estimate uncertainties.

2.2 Creation of the computational model

A schematic of the proposed model is presented in Fig. 1. The problem will be divided into two parts, the initial solid thermal model from which the material layer and contact resistances will be verified and validated, and the CHT part in which we will expand the model to include gaz and convective heat transfer.

2.2.1 Geometry

The geometry of the component will be provided by Plane&Simple in the form of a CAD-file. The CAD might have to go through a cleaning step depending on the geometrical features involved, assembly parts as well as software tolerances and file types. As much as possible, features that affect the heat flow will be kept and some features may be simplified. By simplifying unnecessary CAD-features, meshing and simulation will be easier and faster reducing use of computational

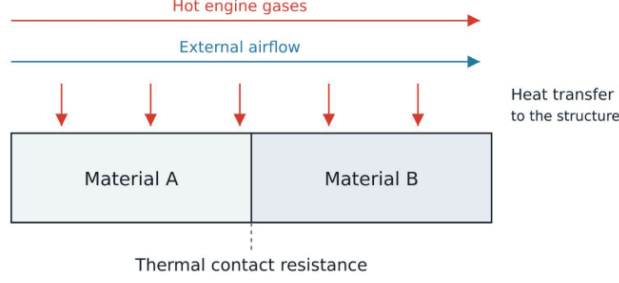


Figure 1: Schematic of the computational model.

resources. The section of the plane that is of interest is shown in Fig. 2. Red dots could represent the thermocouple measuring locations. The same locations will be used for the thermocouples in the experiments and the model to enable reliable comparisons.

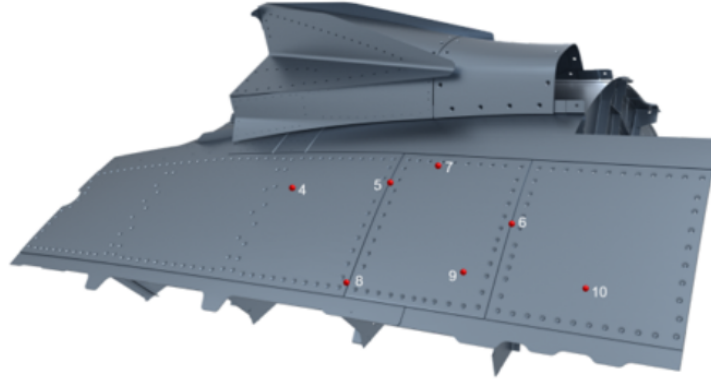


Figure 2: Aircraft section of interest

2.2.2 Creation of a solid thermal model

One create a computational model to predict the temperature fields related to conduction. This solid model will reproduce the conditions of the experimental tests and be used as the first part of the final CHT model. To each material, one will assign its thermal conductivity, density and specific heat capacity. The simulations will reproduce the initial temperatures and heating conditions of the lab tests (accounting for heat losses to the surroundings and supports)([Ansys, Inc., 2025b](#), Sections 17.2.1 and 8.4.15.5). Contact resistances will be introduced to describe the join between the assembled parts. Calculated temperatures will be extracted at the thermocouple locations and measurement times for comparison with the experimental data.

2.2.3 Identification of thermal contact resistances

Contact resistances will be determined using the tests selected for calibration. In practice, the solid-body simulations will need to be rerun while adjusting these resistances to minimize the discrepancies between calculated and measured temperatures. One can also use a weighted least-squares criterion to assign less weight to measurements with higher uncertainty [NIST/SEMATECH](#). The optimization process will rely on physically plausible parameter ranges and multiple initial estimates. A preliminary sensitivity study can also be conducted to verify whether the temperature measurements allow the effects of the various resistances to be distinguished and, so, accurately characterized. In cases where this is not possible, physically comparable contacts may be grouped

together. Before adjusting the contact resistances, one will check mesh sensitivity, time-step sensitivity where relevant, and numerical convergence. This will reduce the risk of adjusting the resistances to compensate for inaccuracies in the numerical solution [NASA Glenn Research Center \(2021b\)](#).

2.3 Development of the CHT model

2.3.1 Addition of conjugate heat transfer

Conjugate heat transfer or CHT is a way to calculate both solid heat transfer and fluid heat transfer at the same time. This can be accomplished by coupling the heat transfer at the interface between solid and fluid. This requires one fluid domain to solve continuity and momentum equations, and one solid domain to solve heat conduction. CHT can be 1-way coupled or 2-way coupled.

In 1-way coupling the domain of one type of heat transfer is solved first, for example the solid. In this way the heating condition at the interface is established. The condition at the wall interface is then utilized as boundary value for the other domain, in this case the fluid. The heat transfer is only calculated in one direction from one domain to the next.

In a 2-way coupling the heating conditions at the interface are continually updated ([Ansys, Inc., 2025b](#), Section 17.2.2.6). The temperature and heat flux will be solved together rather than independently. The hull temperature is expected to have an effect on the heat transfer back to the engine exhaust gas. The plan for this project is thus to employ a 2-way coupling approach, to ensure that the solid-fluid interaction is properly represented.

For this project, the solid model will first be validated by comparison with reference laboratory tests (using an experimental dataset set aside specifically for this purpose) before being used in the conjugate heat transfer (CHT) calculation. Next, a local gas domain will be added around the relevant aircraft surfaces. The full flow fields available for the aircraft will be used to define the boundary conditions for this domain, supplemented as necessary by data on exhaust gas temperature, flow rate, and composition. The heating conditions applied in the laboratory will be replaced by operational conditions and thermal conditions on the remaining solid surfaces will be defined based on the actual installation configuration. The previously determined contact resistances will be reused in the CHT model, provided that the joints between components remain sufficiently similar to the cases for which those resistances were originally determined.

2.3.2 Comparison of simulation approaches

Three configurations will be compared: a steady one-way calculation, a steady two-way calculation and an unsteady two-way calculation. One can retain two-way coupling for the reference approach described above. For the one-way comparison, the gas calculation should probably use a specified reference wall temperature. The resulting heat flux will be transferred to the solid model without feedback from the calculated solid temperature. Plus, the reference temperature will be documented and varied to assess its influence.

The steady and unsteady calculations will initially use the same SST k - ω turbulence model and operating conditions ([Ansys, Inc., 2025a](#), Section 4.4.3). Steady RANS will seek a time-independent mean solution whereas the use of URANS will follow its time evolution while still modelling turbulent fluctuations ([Ansys, Inc., 2025a](#), Section 4.1.1). Comparisons have to cover structural temperatures, wall heat fluxes, total heat-transfer rates and computational cost. Unsteady averages and peaks have to be evaluated after the initial transient over a sufficiently long observation period.

2.4 Verification & Validation

Verification is a common term for making sure the numerical equations sufficiently solve the modeled problem. Both for the solid thermal model and the CHT model, a mesh study will be performed to verify the discretizations errors and guarantee the results are independent of mesh size and quality (Gasparini, Simularge). The mesh analysis will be done according to a process outlined by Celik et al. (2008) where three grids are compared and errors are estimated for temperatures and heat flux.

In addition to the mesh study, the time step must also be verified for the unsteady simulation. This will be done by consecutively reducing time step size and calculating the relative percent errors E according to :

$$E = \left| \frac{\phi_{\text{fine}} - \phi_{\text{coarse}}}{\phi_{\text{fine}}} \right| \cdot 100 \quad (1)$$

Where ϕ_{fine} represents the solution when using smaller time step size, and ϕ_{coarse} represents the solution when using larger time step size. All subsequent simulations will utilize a time step and grid size where the error is sufficiently small.

It is also necessary to verify that the domain dimensions do not artificially alter the calculated heating. This is done by shifting boundaries or enlarging the gas volume while maintaining mesh resolution near the structure.

Convergence will also be checked using equation residuals and monitored temperatures and heat-transfer rates. One will examine the overall energy balance (including energy storage in transient calculations). These checks can complement the mesh, time-step and domain-size studies NASA Glenn Research Center (2021b). Validation will be done by comparing the model output with experimental results. Therefore it is of utmost importance that the thermocouples share placements on the surface between experiment and numerical model. A python code will be written to plot temperatures and heat flux for comparison.

For the solid model, this comparison will be based on an experimental test excluded from the contact resistance calibration phase. The TCR values obtained from the first test are retained, while the conditions of the second test (the one designated for validation) are simulated. Agreement can be assessed based on temperature histories, measurement uncertainty, and estimated numerical uncertainty, in accordance with pre-defined acceptance criteria NASA Glenn Research Center (2021a). If the results require modifications to the model and resistance values, a new independent test will be required for a subsequent evaluation. Note that, since thermocouples measure temperature, heat flux predictions cannot be validated directly from these measurements alone. The laboratory tests will therefore assess how well the solid model reproduces the measured temperatures under electrical heating. They will not directly validate the heat transfer between the engine exhaust and the structure, since this flow is not reproduced in the laboratory. NASA Glenn Research Center (2021a).

2.5 Physical assumptions and sensitivity analysis

The operating data will be used to assess the need for temperature-dependent properties, compressibility and eventually radiative heat transfer (Ansys, Inc., 2025a, Sections 5.2.1 and 5.3.10). The selected assumptions have to be documented. A sensitivity analysis will then vary the contact resistances, material properties and exhaust boundary conditions within ranges that have to be justified. The resulting changes in temperatures and heat-transfer rates will identify the most influential inputs and help determine whether differences between the simulation approaches remain meaningful when these uncertainties are considered.

3 Schedule

The project is planned on a 20 week period (including required tests, calculations and deliverables). The work can be first divided into a Work Breakdown Structure (WBS). Then one can represent tasks dependencies in a network chart. This network serves as the basis for the Gantt chart.

3.1 WBS

The WBS 5 contains three levels : the project itself, 6 work packages and 20 tasks (T01 through T20). The work packages cover project management, testing, the solid model, the CHT model, transfer to Plane& Simple, reports, presentations. The 3rd level tasks are incorporated into the network using the same ID. The WBS describes the work content and the network chart specifies the execution sequence.

3.2 Dependencies and critical path

The network 6 specifies the results required to start each task. For example, the characterization of the TCR depends on the test measurements and the solid model. The CHT calculation depends on the validation of the solid model and the preparation of the calculation of the gas-phase. So, certain activities (like testing and solid model construction) can proceed in parallel. In the schedule, the interim presentation also serves as a milestone before the comparison of coupling methods.

The network results in a 20 week timeline and it is based on estimated durations. Two critical branches (T06-T09 and T18-T19) each span 3 weeks following T05 and converge at T10. A delay in a critical task pushes back the project completion date unless adjustments are made. T20 covers monitoring and documentation distributed across the entire project.

Note : carrying out parallel activities assumes the availability of the necessary personnel and computing resources.

3.3 Gantt chart

The 20-week plan below distributes the work across the Work Breakdown Structure (WBS) shown in Figure 5 (WP1–WP6) and the Network Chart shown in Figure 6 at the appendix, together with the milestones and gates.

The project timeline reflects two critical branches that converge at week 10: one through solid model validation (T06 → T09 → T10) and another through gas flow simulation (T08 → T09 → T10). Any delay in these critical tasks will directly impact the project completion date. The critical path continues through T11 → T12 → T15 → T16 → T17, with zero schedule slack on final deliverables.

Five progress gates (G1–G5) are scheduled to verify project progress and allow for corrective action. Gate G1 (week 2) confirms experimental protocol approval; G2 (week 4) validates measurement quality; G3 (week 8) confirms solid model validation within ± 1 °C of thermocouple measurements; G4 (week 12) verifies mesh convergence and coupling methodology; and G5 (week 15) freezes results after sensitivity analysis to protect writing time.

The experimental campaign (T02–T03) is scheduled in weeks 1–5 to align with facility availability. The solid thermal model development (T04–T05) overlaps with gas-phase preparation (T07) to exploit parallel work. A single rerun opportunity is reserved in week 5 should measurement quality be insufficient. If transient simulations prove computationally prohibitive, the project will prioritize steady-state results and restrict transient analysis to representative cases. Off-site work concludes at the end of week 17.

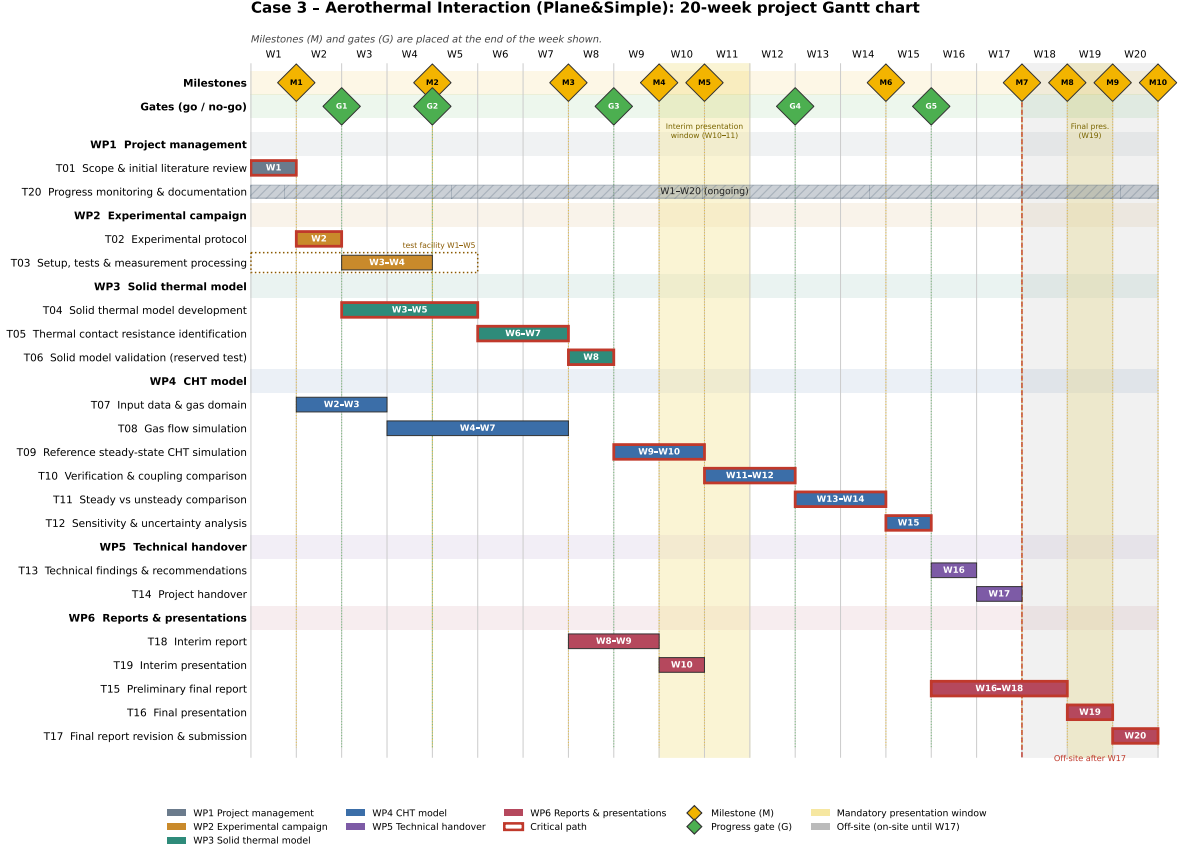


Figure 3: 20-week Project Gantt Chart

4 Risk Assessment

The proposed project combines experimental testing, Thermal Contact Resistance (TCR) calibration, and advanced Conjugate Heat Transfer (CHT) simulations. As these activities are strongly interconnected, uncertainties arising in one phase may propagate throughout the project and affect both the schedule and the quality of the final results. A risk assessment was therefore conducted to identify critical activities, evaluate their potential impact, and define mitigation strategies.

4.1 Experimental Uncertainties

The first major source of uncertainty is associated with the experimental campaign performed on the aircraft section. The measurements obtained from thermocouples will be used to calibrate and validate the numerical model. Consequently, any measurement errors caused by sensor inaccuracies, imperfect thermocouple positioning, heat losses to the surroundings, or variations in the electrical heating system may directly affect the reliability of the estimated TCR values. Since the TCR calibration forms the basis of all subsequent simulations, inaccurate experimental data could compromise the credibility of the entire modelling methodology.

Another significant risk is related to the limited availability of the experimental facility. The test campaign must be completed within a restricted time window during the first weeks of the project. Any delays in test preparation, equipment malfunction, or the need for additional measurements

could postpone subsequent activities such as data analysis, model calibration, and validation.

4.2 Numerical Modelling Risks

The development of a reliable CHT model represents one of the main technical challenges of the project. The aircraft structure consists of multiple materials connected through interfaces where the thermal contact resistance is initially unknown. Determining representative TCR values may require several calibration iterations and sensitivity studies, increasing both computational effort and project duration.

Additional uncertainty is associated with the selection of turbulence modelling approaches. Different methods, such as steady-state RANS and transient URANS simulations, may predict different thermal loads on the structure. Selecting an inappropriate modelling approach could lead to inaccurate predictions and require further validation work.

Mesh generation also represents a potential risk. Accurate prediction of heat transfer near material interfaces and in regions exposed to hot exhaust gases may require highly refined meshes. This can significantly increase computational costs and simulation times, potentially affecting the project schedule.

4.3 Computational and Resource Risks

Although access to a high-performance computing cluster with 320 cores is available, CHT simulations remain computationally demanding. Long computation times, convergence difficulties, numerical instabilities, or unexpected software issues may delay project progress. This risk is particularly important for transient simulations, which generally require substantially more computational resources than steady-state calculations.

The project also depends on external flow-field data previously generated by the company. These data are intended to provide realistic boundary conditions for the aircraft section simulations. Delays in obtaining these datasets, or limitations in their quality or completeness, could require modifications to the modelling strategy and introduce additional uncertainty.

4.4 Critical Activities and Potential Impact on the Project

Several activities identified in the Gantt chart are located on the project’s critical path and therefore represent the highest-risk tasks.

The most critical activity is the experimental characterization of the aircraft section and the determination of the Thermal Contact Resistance. If this task cannot be completed successfully, the numerical model cannot be properly calibrated, which would undermine the reliability of all subsequent simulations.

The second critical activity is the validation of the numerical model against experimental measurements. Since one of the project’s primary objectives is to develop a reliable modelling methodology, poor agreement between simulations and experiments would significantly reduce confidence in the final results. Additional calibration loops and sensitivity studies may then be required, leading to schedule overruns.

Finally, the development of the final CHT model is dependent on the successful completion of all previous tasks. Delays occurring during experiments, calibration, or validation may therefore prop-

agate throughout the remainder of the project.

4.5 Contingency Plans (Plan B)

To reduce the impact of these risks, several contingency strategies are included in the project plan.

If experimental measurements cannot be completed as originally planned, literature values and engineering estimates for thermal contact resistance may be used to continue the CFD development. Although this would reduce the quality of the validation process, it would still allow the investigation of the proposed CHT methodology.

If transient simulations prove too computationally expensive or experience convergence difficulties, the study can prioritize steady-state simulations and restrict transient analyses to a limited number of representative operating conditions.

Similarly, if advanced turbulence models cannot be successfully implemented within the available timeframe, validated RANS approaches may be adopted as a fallback solution to ensure completion of the project objectives. The largest uncertainty of the project concerns the successful validation

Gantt Activity	Potential Failure Mode	Consequence	Severity	Mitigation Strategy
Experimental preparation	Equipment availability issues or preparation delays	Postponement of validation activities	High	Early planning and preparation of test procedures
Experimental measurements	Sensor failure or inaccurate temperature data	Incorrect TCR estimation	High	Sensor calibration and repeated measurements
TCR calibration	No unique or representative TCR value identified	Reduced model credibility	High	Sensitivity studies and optimization-based calibration
Mesh generation	Excessive mesh size	Long simulation times	Medium	Progressive mesh refinement strategy
CFD simulations	Numerical instability or non-convergence	Delayed project completion	High	Begin with simplified models before increasing complexity
Model validation	Poor agreement between simulations and experiments	Additional calibration iterations required	Very High	Schedule contingency time and uncertainty analysis
Final CHT simulations	Excessive computational cost	Reduced number of investigated cases	Medium	Prioritize the most relevant operating conditions

Figure 4: FMEA-Style Risk Assessment

of the numerical methodology. Since the final objective is not only to perform simulations but also to establish a reliable predictive framework for aerothermal analysis of aircraft structures, any issue affecting the experimental campaign, TCR determination, or validation phase will have a disproportionately large impact on the final outcome. For this reason, additional buffer time should be allocated around these activities in the Gantt chart, and regular verification checkpoints are implemented throughout the project to minimize the likelihood of major schedule deviations.

5 Conclusion

This planning report has outlined the methodology, schedule, and risk management strategy for developing a validated CHT model of the engine exhaust–fuselage interaction for Plane Simple AB. The proposed approach — progressing from a calibrated solid thermal model to a fully coupled CHT simulation, supported by a structured verification and validation process — provides a robust framework for meeting the project’s objectives within the 20-week timeline.

Beyond this project’s scope, the methodology could later be extended to other fuselage regions or engine configurations, and combined with thermomechanical analysis to assess resulting structural stresses.

6 Appendix

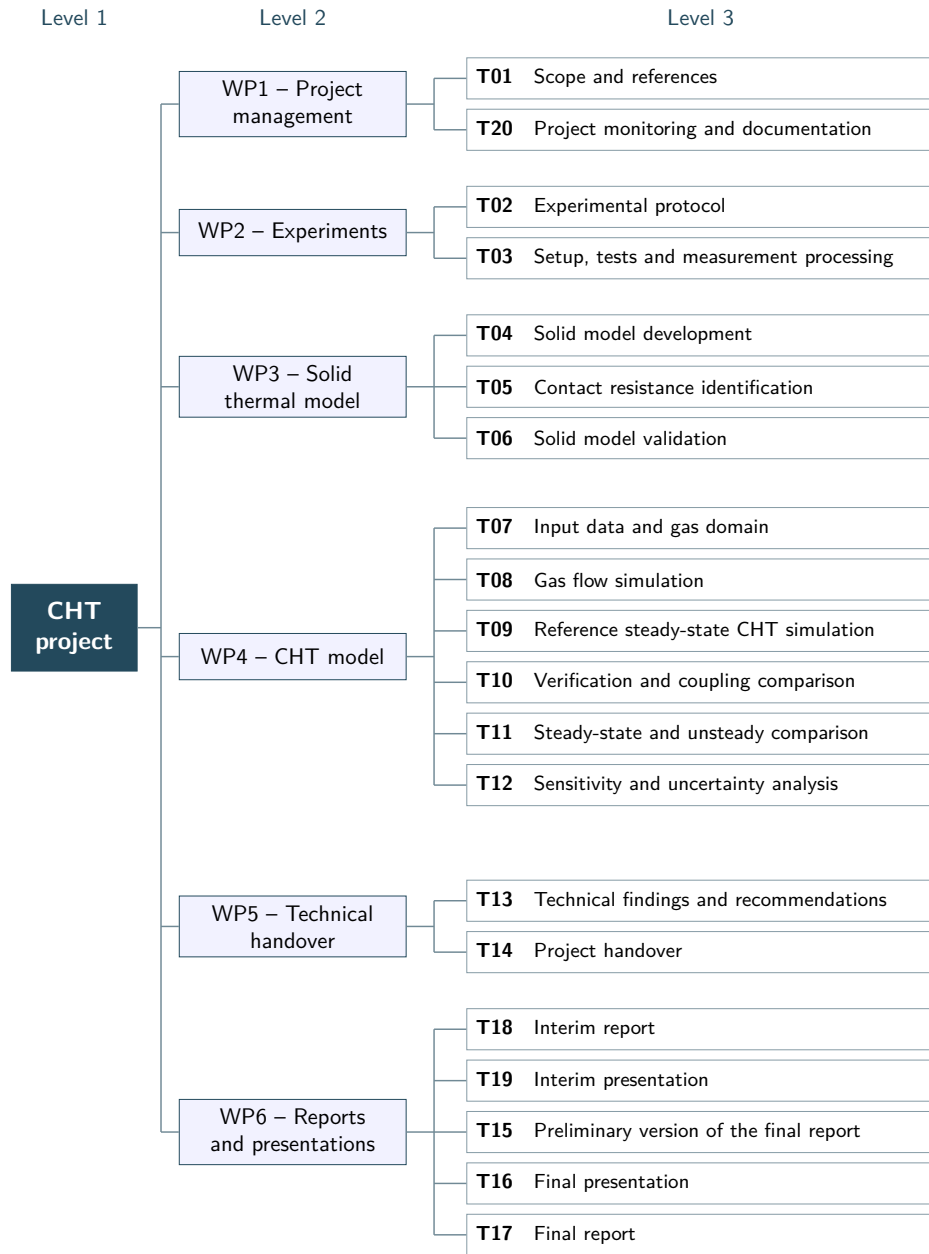


Figure 5: WBS in 3 levels. WP : WorkPackage ; T : Task

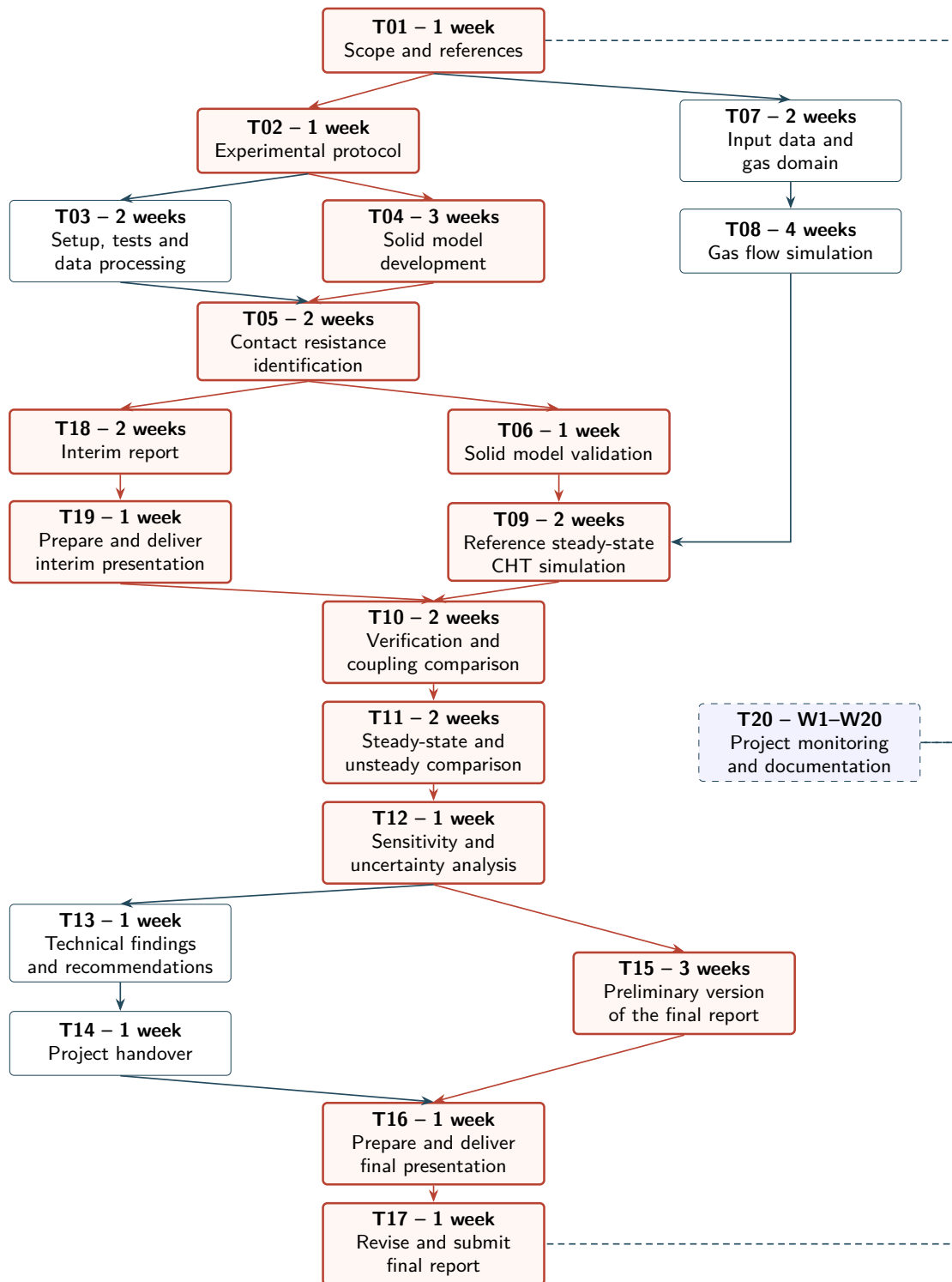


Figure 6: Network chart showing estimated durations - Arrows indicate dependencies and red boxes highlight critical tasks. Dotted lines connect the continuous activity T20 to the start and end of the project without adding a dependency.

Author Contribution Statement

J.Y. handled the introduction.

P.D. and J.Y. prepared the methodology section.

P.D. created the WBS and the network chart.

M.S. produced the Gantt chart.

N.D. prepared the risk assessment section and the conclusion.

The entire team challenged the choices made and validated both the content and the format of the report.

Author Declaration of the Use of Generative AI

During the preparation of this work, the team used AI agents to shed light on certain physical concepts, to help refine English word choices and to assist with the initial literature search.

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